



Microsoft's new flagship

Microsoft is shaking it's surface hardware lineup, rumours say that there's a new flagship coming. <https://digit.in/sep21-61>



Pixel 6 with samsung modem

Pixel 6 may sport a 50-megapixel camera, 5G modem from Samsung. <https://digit.in/sep21-62>

NVIDIA GeForce RTX 3070 Ti Graphics Card

An odd predicament



NVIDIA had announced the GeForce RTX 3070 Ti alongside the RTX 3080 Ti at COMPUTEX 2021. When we reviewed the RTX 3080 Ti, we'd found it to be very close to the RTX 3090 in terms of performance. So it's completely natural for folks to assume that the RTX 3070 Ti might end up a great performer as well. Would it match up to the RTX 3080 in terms of performance? Not quite.

NVIDIA GEFORCE RTX 3070 TI SPECIFICATIONS

The RTX 3070 Ti, like the RTX 3080 Ti uses GDDR6X memory which offers a decent improvement in bandwidth over the GDDR6 memory that the RTX 3070 uses. However, the bus width remains the same at 256-bit wide. This shouldn't be seen as a bottleneck since there's plenty of headroom and the 3070 Ti has a max theoretical memory bandwidth of 608.3 GB/s whereas the RTX 3070 had a max bandwidth of 448 GB/s. The core boost clock is also higher on the RTX 3070 Ti by about 45 MHz while the base clock is higher by 75 MHz. This leaves us with the core configuration. That's where the RTX 3070 Ti falls short of impressing us. Compared to the RTX 3070, the RTX 3070 Ti only has two additional SMs



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which translates to 256 additional CUDA cores. In an entry level segment, a 2 SM increment is a significant upgrade as long as the clock speeds and memory bandwidth are sufficient. In a mid-range or even high-end segment, a 2 SM increment is barely worth anything.

NVIDIA GEFORCE RTX 3070 TI PERFORMANCE

The NVIDIA RTX 3070 Ti scores about 14,416 in TimeSpy which puts it ahead of the RTX 3070's 12,618 by 14 per cent which is a nice delta. Similarly, in Fire Strike Ultra, the 3070 Ti scores a little over 10 per cent higher than the RTX 3070. TimeSpy scores are at par with the AMD RX 6800 and Fire Strike Ultra scores are about 9-10 per cent higher. So far, it's looking good in the synthetics. The RTX 3070 Ti completes the Blender benchmark about 6 seconds faster than the RTX 3070 which is a decent improvement. With a much higher memory bandwidth, the RTX 3070 Ti scores way ahead of the RTX 3070 in IndigoBench.

In Borderlands 3, the RTX 3070 Ti scores about the same as the RTX 3070. At higher resolutions, you can see a difference of about 4-5 FPS. In Death Stranding, we see about a 5 per cent improvement in 1440p and 4K resolu-

tions. That's literally as many in FPS. In Witcher 3, we see a performance improvement of 1-4 per cent at 1080p and 1440p with 4K gaming showing a much higher 15 per cent improvement.

NVIDIA GEFORCE RTX 3070 TI RAY TRACING

NVIDIA RTX graphics cards have continued to score higher in synthetic ray-tracing benchmarks such as 3D Mark Port Royale with the RTX 3070 Ti scoring 6 per cent higher than the RTX 3070 and about 15 per cent higher than the AMD Radeon RX 6800 XT.

PRICING AND AVAILABILITY

The NVIDIA GeForce RTX 3070 Ti has an official price of ₹53,500 for the Founders Edition graphics cards while the partner cards cost about ₹1,20,000 in India owing to the GPU shortage issue. Industry reports had indicated that GPU prices would have normalised around mid-September and we did see a significant price drop towards the end of July. However, we can no longer blame the global chip shortage for the high prices since practically every major computer parts store has been putting out plenty of units for sale.

VERDICT

The NVIDIA GeForce RTX 3070 Ti puts gamers in an odd predicament. There's barely any performance to be gained with the RTX 3070 Ti since it's practically similar to the RTX 3070's performance but has a much higher price tag. You'd be shelling about 10K for 1-5 per cent additional performance but in reality, you're paying 20-35K. There's no way gamers can make any sense of this price-to-performance factor but if you happen to be a content creator, then the 3070 Ti is indeed worth the additional 10K.

—Mithun Mohandas

SPECIFICATIONS

STREAM PROCESSORS: 6144 | CU: 48 | BOOST CLOCK: 1770 MHz | VRAM: 8 GB GDDR6X | BUS-WIDTH: 256-bit | CONNECTORS: 1x 12-PIN | TBP: 290W

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